

Network Scanning with Nmap

Objective: Learn how to scan a network and identify open ports.

Performing different types of scans such as:

- **Basic scan: nmap <IP>**

```
C:\Users\VICTUS> nmap 192.168
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-28 20:07 India Standard Time
Nmap scan report for 192.168
Host is up (0.00092s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 0.57 seconds
```

- **Service detection: nmap -sV <IP>**

```
C:\Users\VICTUS> nmap -sV 192.168
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-28 20:05 India Standard Time
Nmap scan report for 192.168
Host is up (0.00044s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
139/tcp    open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.11 seconds
```

- **Aggressive scan: nmap -A <IP>**

```
C:\Users\VICTUS> nmap -A 192.168
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-28 19:56 India Standard Time
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 97.42% done; ETC: 19:57 (0:00:00 remaining)
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 97.42% done; ETC: 19:57 (0:00:00 remaining)
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 97.42% done; ETC: 19:57 (0:00:00 remaining)
Packet Tracing enabled.
NSE: UDP 192.168.56.1:51006 > 192.168.445 | 00000000: 13 37 00 00 00 01 00 00 00 00 00 20 43 4b 41 7
CKA
00000010: 41 41 41 41 41 41 41 41 41 41 41 41 41 41 41 41 AAAAAAAAAAAAAAAAAA
00000020: 41 41 41 41 41 41 41 41 41 41 41 41 41 00 00 21 AAAAAAAAAAAAAAAA !
00000030: 00 01

NSOCK INFO [12.1600s] nssock_trace_handler_callback(): Callback: READ SUCCESS for EID 346 [192.168.445] (240 bytes)
NSE: TCP 192.168.54509 < 192.168.445 | 00000000: 00 00 00 ec fe 53 4d 42 40 00 00 00 00 00 00 00 00 SM
00
00000010: 00 00 01 00 01 00 00 00 00 00 00 00 00 00 00 00
00000020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000030: 00 00 00 00 31 32 33 34 35 36 37 38 39 30 31 32 123456789012
00000040: 33 34 35 36 41 00 01 00 11 03 02 00 e7 ca 9a 34 3456A 4
00000050: a9 44 8a 4c a0 22 20 ad cf cd f7 92 07 00 00 00 00 D L "
00000060: 00 00 80 00 00 00 80 00 00 00 80 00 8f df af 7e ~
00000070: ed 9f db 01 00 00 00 00 00 00 00 00 00 00 2a 00 *
00000080: b0 00 00 00 60 28 06 06 2b 06 01 05 05 02 a0 1e '( +
00000090: 30 1c a0 1a 30 18 06 0a 2b 06 01 04 01 82 37 02 0 + + 7
000000a0: 02 1e 06 0a 2b 06 01 04 01 82 37 02 02 0a 00 00 + 7
000000b0: 00 00 00 00 01 00 26 00 00 00 00 00 01 00 20 00 &
000000c0: 01 00 e1 ab 40 4c a3 68 79 cf b0 30 b6 10 1b 29 @L hy 0 )
000000d0: 41 9d 1b ec 8a 83 71 e8 6c f7 04 c9 e4 b5 e1 6d A q l m
000000e0: 7a 8a 00 00 02 00 04 00 00 00 00 00 01 00 02 00 z

NSOCK INFO [12.600s] nssock_trace_handler_callback(): Callback: WRITE SUCCESS for EID 355 [192.168.56.1:137]
NSE: UDP 192.168.56.1:51006 > 192.168.445 | SEND
NSOCK INFO [12.220s] nssock_readbytes(): Read request for 1 bytes from IOD #15 [192.168.445] ID 362
NSE: TCP 192.168.56.1:54509 > 192.168.445 | CLOSE
NSOCK INFO [12.2220s] nssock_iodelete(): nssock_iodelete (IOD #14)
NSOCK INFO [13.0460s] nssock_trace_handler_callback(): Callback: READ TIMEOUT for EID 314 [192.168.445]
NSE: UDP 192.168.56.1:51005 > 192.168.445 | CLOSE
NSOCK INFO [13.0930s] nssock_iodelete(): nssock_iodelete (IOD #13)
NSOCK INFO [13.2320s] nssock_trace_handler_callback(): Callback: READ TIMEOUT for EID 362 [192.168.445]
NSE: UDP 192.168.51006 > 192.168.445 |137 | CLOSE
NSOCK INFO [13.2780s] nssock_iodelete(): nssock_iodelete (IOD #15)
NSOCK INFO [13.2780s] nssock_iodelete(): nssock_iodelete (IOD #16)
NSOCK INFO [13.2810s] nssock_connect_tcp(): TCP connection requested to 192.168.445 (IOD #16) EID 368
NSOCK INFO [13.2820s] nssock_trace_handler_callback(): Callback: CONNECT SUCCESS for EID 368 [192.168.445]
NSE: TCP 192.168.51006 > 192.168.445 | CONNECT
NSOCK INFO [13.3410s] nssock_iodelete(): nssock_iodelete (IOD #17)
NSOCK INFO [13.3430s] nssock_connect_udp(): UDP connection requested to 192.168.445 (IOD #17) EID 376
NSE: TCP 192.168.51006 > 192.168.445 | 00000000: 00 00 00 31 ff 53 4d 42 72 00 00 00 00 18 45 60 1 SM
Br E'
00000010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 e9 56 V
00000020: 00 00 01 00 00 0e 00 02 4e 54 20 4c 4d 20 30 2e NT LM 0.
00000030: 31 32 00 02 00 12
```